



Guidelines

Hypokalaemia

HRM013

Scope

Site	Service/Department/Unit	Disciplines
SCGH	All Areas	Medical, Nursing, Pharmacy

Disclaimer

These guidelines have been produced to guide clinical decision making for the medical, nursing and allied health staff of Sir Charles Gairdner Hospital. They are not strict protocols, and **they do not replace the judgement of a senior clinician**. Clinical common-sense should be applied at all times. These clinical guidelines should never be relied on as a substitute for proper assessment with respect to the particular circumstances of each case and the needs of each patient. Clinicians should also consider the local skill level available and their local area policies before following any guideline.

Introduction

Potassium is the major intracellular cation and is mainly found in skeletal muscle; only 2% of total body potassium is in extracellular fluid.

Potassium is essential for maintenance of:

- Electro-potential gradient of the cell membrane
- Isotonicity
- Acid-base balance

The kidney has a major role in potassium homeostasis.

PathWest reference range for plasma potassium is 3.5 to 5.2mmol/L.

Clinical Presentation¹⁻⁵

Signs and symptoms of hypokalaemia are late and non-specific including:

- Fatigue and general weakness
- Constipation
- Weakness
- Cramps
- Paraesthesia
- Muscle paralysis (late sign)
- Arrhythmias (SVT, VT and Torsades de Pointes)
- ECG changes (ST depression, 'T' wave flattening, 'U' waves)

It is essential to identify and minimise any underlying cause(s) when treating hypokalaemia:

Common Causes and Precipitating Factors

Potassium transcellular shift (redistribution of potassium ions into cells)	• Drugs – Insulin/glucose therapy, beta-adrenergic stimulant (salbutamol), catecholamines (adrenaline/epinephrine), xanthines (theophylline, caffeine), verapamil, sodium bicarbonate and chloroquine intoxication • Metabolic alkalosis • Hypokalaemic periodic paralysis • Anabolic state
Increased potassium loss	• Drugs – diuretics (frusemide/furosemide, thiazides, chlorthalidone, acetazolamide, mannitol), laxative abuse (phenolphthalein), liquorice, corticosteroids (fludrocortisone, prednisolone), antibiotics (penicillins) • GI losses – diarrhoea, vomiting, ileostomy, intestinal fistula, villous adenoma • Renal – renal tubular disorders, Bartter syndrome, Liddle's syndrome, Gitelman syndrome, nephrogenic diabetes insipidus • Endocrine – hyperaldosteronism, Cushing's syndrome, Conn's syndrome, severe ectopic adrenocorticotrophic hormone syndrome • Dialysis (haemodialysis / peritoneal dialysis), plasmapheresis, • Haemodilution
Inadequate potassium intake	• Poor dietary intake, refeeding syndrome
Magnesium deficiency (increases renal potassium loss)	• Poor dietary intake, refeeding syndrome • Increased magnesium loss e.g., drugs such as aminoglycosides, cisplatin, foscarnet, amphotericin

Management Strategies⁶⁻¹⁰

Step 1

Assess clinical symptoms, co-morbidities (e.g. arrhythmias with digitalis use), plasma potassium level and severity of hypokalaemia to exclude whether emergency measures are indicated.

Step 2

Replace potassium accordingly with close monitoring:

Severity	Treatment	Comments
MILD 3.0 to 3.4mmol/L (~150mmol deficit)	Asymptomatic ☞ Potassium Supplements: Oral Potassium Effervescent 548mg Tablets 2 tablets BD (= 56mmol/day) or Potassium Chloride 10% Oral Liquid 15mL TDS (= 60mmol/day) or Potassium Chloride SR 600mg Tablets 2 tablets TDS (= 48mmol/day)	• Usually asymptomatic, except for patients with heart condition or cirrhosis. • Consider IV ONLY if patient cannot tolerate oral. • Monitor K⁺ level daily and adjust dose accordingly. • Continue treatment until K⁺ is within range for about 3 days.
	Symptomatic – see SEVERE No or Minor Symptoms ☞ Potassium Supplements: Oral Potassium Effervescent 548mg Tablets 2 tablets TDS (= 84mmol/day) or Potassium Chloride 10% Oral Liquid 25mL TDS (= 100mmol/day) or Potassium Chloride SR 600mg Tablets 3 tablets QID (= 96mmol/day) Alternatively, ☞ Potassium Supplements: Intravenous Potassium Chloride IV Infusion 20 to 40mmol – REPEAT as required	• <u>Oral route is preferred</u> However, IV potassium may be used as an adjunct to oral if large doses are required, as these can cause gastric irritation. • Check K⁺ level after every 20 to 40mmol IV given and at least once daily for oral potassium. • Adjust treatment according to plasma K⁺ level.
	Symptomatic ☞ Potassium Supplements: Intravenous Potassium Chloride IV Infusion 40mmol – REPEAT as required Maximum Recommended: 2 to 3mmol potassium/kg/24hrs up to 400mmol potassium/24hrs	• Check K⁺ level after every 40 to 80mmol and at least once daily to determine the need for further infusion - to avoid hyperkalaemia. • Oral potassium (maximum dose as in MODERATE) can be used as adjunct if patient able to tolerate.

With normal acid–base status, approximately a decrease of 1mmol/L in plasma concentration represents a whole-body potassium deficit of 200 to 300mmol, while a 2mmol/L decrease is associated with a deficit of 500 to 600mmol/L. Alternatively, an initial total potassium deficit in mmol can be estimated:

$$\text{Total K}^+_{\text{deficit}} (\text{mmol}) = [(K^+_{\text{target}} - K^+_{\text{measured}}) \times \text{body weight (kg)} \times 0.4] + [1.25\text{mmol/kg} \times \text{body weight (kg)}]$$

Step 3

Assess renal function, acid-base status and underlying cardiac disease.

- Renal impairment – extreme caution is required as these patients are prone to severe life-threatening hyperkalaemia.
- Potassium deficit is greater in metabolic acidosis than metabolic alkalosis:
Metabolic Acidosis (pH < 7.35): + 1.0mmol/L K⁺ per 0.1unit pH change

Metabolic Alkalosis (pH > 7.45): - 0.25mmol/L K⁺ per 0.1unit pH change

- Diabetic Ketoacidosis – involves immediate correction and close monitoring of hyperglycaemia with insulin, acid-base balance, fluid loss with intravenous fluids and electrolyte disturbance, particularly potassium loss. All of which will require an individualised care plan.

Step 4

Treat any co-existing hypomagnesaemia and hypocalcaemia.

Step 5

- Evaluate cause(s) of hypokalaemia and address all possible precipitating factors of ongoing potassium loss, such as drugs, vomiting, diarrhoea, pancreatic / small bowel fistula (refer to section Common Causes and Precipitating Factors).
- If hypokalaemia is caused by redistribution of potassium ions into cells, this should be managed rather than giving more potassium as this can cause rebound hyperkalaemia. Investigate possible underlying endocrine or renal abnormalities if suspected.

Practical Points^{1,9}

Considerations- Special Patient Populations

- **Cardiac patients** – maintenance of ≥ 4.0 mmol/L as optimal plasma potassium levels is essential for patients with hypertension, congestive cardiac failure and arrhythmias
- **Diabetic Ketoacidosis** – this is extremely complex and expert advice and management in a high intensity area is almost always required
- **Periodic paralysis or insulin excess** – rebound hyperkalaemia is also a danger when hypokalaemia is due to acute potassium shift into cells.
- **Low bodyweight patients** – extreme caution when replacing potassium, as their low muscle mass increases the chance of overcorrecting the deficit, leading to hyperkalaemia.
- **Severe metabolic alkalosis** (e.g. from vomiting acid stomach contents) –potassium repletion may not be possible until the chloride deficit is corrected.
- **Proven or suspected diuretic drug abuse** – suddenly stopping a high-dose loop diuretic can cause marked fluid retention. Diuretic dosage should be reduced gradually, while restricting sodium and water intake.
- **Pre-operative patients** – it is important to maintain potassium plasma level ≥ 3.0 mmol/L to assure optimal surgical outcome.
- **Fixed mineralocorticoid excess** (e.g. in primary hyperaldosteronism) – restricting sodium intake is also key in achieving potassium repletion

Note: Potassium-sparing drugs (e.g. spironolactone and amiloride) have no place in managing acute hypokalaemia but can be useful for maintenance of potassium level in some cases of chronic hypokalaemia

Prescribing Potassium Supplements

Hypokalaemia may be corrected with potassium supplementation either orally and / or parenterally:

- Oral is the preferred route of administration for potassium replacement and especially with chronic potassium deficit. It is also easy to administer and readily absorbed from the gastro-intestinal tract.

- IV potassium chloride should only be prescribed when the oral or enteral route is unavailable or clinically inappropriate.
- Generic name should be used when prescribing, avoid brand names (i.e. Slow K) as well as chemical abbreviations (i.e. KCl, K⁺).
- IV potassium must be prescribed in “mmol” of potassium, and must specify the dose, fluid, volume, and rate of administration.
- IV potassium chloride must be prescribed on SCGH intravenous fluid order / fluid therapy chart (ENTER), NOT on WA Hospital Medication Chart (WA HMC).
- The name of the consultant providing the approval must be documented on SCGH intravenous fluid order / therapy chart (i.e. Approved by Dr....).

If prescribing IV potassium outside the guidelines, the order should be endorsed by the treating team consultant.

Monitoring

- Plasma potassium concentration – frequently at least once daily.
- Recheck plasma potassium level:
 - ☒ 4 hours after last ORAL dose
 - ☒ 1 hour after IV INFUSION complete
- Electrocardiogram (ECG) – especially patients with a pre-existing cardiac condition, renal impairment and/or when IV potassium is given at rate exceeding 10mmol per hour.
- Fluid balance – when giving large volume intravenously via the peripheral route. This is a major issue during IV potassium replacement (as 1 litre per hour IV fluid may not be appropriate in some clinical settings).
- Monitor U&Es, bicarbonate, chloride and glucose if necessary.
- Plasma magnesium – if level is low, the ability to retain potassium will be compromised

Related Documents

Legislation

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016](#)

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline No.51 Medication Management](#)
- [SCGH Nursing Practice Guideline No.4 Intravenous Therapy](#)
- NSQHS Standards: 1,4,6,8

References

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